UNIVERSIDADE DE AVEIRO

DEPARTAMENTO DE ELECTRÓNICA, TELECOMUNICAÇÕES E INFORMÁTICA

Information and Coding (2026/27)

Lab work n$^{\circ}$ 2 — Due: 17 Dec 2026 — Discussion: 21 Dec 2026

Intro

- You may choose the programming language you find more appropriate to implement the project. You may even use different programming languages in different parts of the project. However, bear in mind that some of the tasks require computationally efficient languages, because running time and memory usage will be a factor of assessment.
- Use a github repository to manage the developement of your software.
- Your software must be provided in a way easy to install and be prepared to run at least in a Linux environment. Appropriate installation mechanisms (for example, makefiles), installation instructions and execution examples (for example, described in a readme file) must be provided.
- You will have some datasets available in eLearning to begin testing and tuning your codecs. Feel free to get more data and to use it. However, be careful not to over-specialize your programs in certain datasets, because we will also run performance tests using a private dataset.

Part I

1. Create a program, named `img_hist`, able to compute and plot histograms of image files in RAW, PGM or PPM format. Image pixels can occupy one byte, i.e., 8 bit per pixel (bpp) in the case of RAW and PGM, two bytes, i.e., 16 bpp, also in the case of RAW and PGM, or three bytes in the case of PPM. If in RAW format, the number of rows, number of columns, number of bits per pixel and, in the case of 16 bpp images, the byte order (big-endian or little-endian), need to be provided. Note that 16 bpp PGM images assume big-endian ordering. PPM images contain information of the three color channels, R, G and B. This program should be prepared to:
    - Plot the histogram of the pixel values. If in PPM format, plot the histograms of the three color channels. Also, calculate and print the corresponding first order entropy of the pixel data in the case of RAW and PGM, or the three values for the R, G and B channels for PPM images. Moreover, it should calculate and print the corresponding entropies of horizontal and vertical differences between neighboring pixels.

- Do the same, but for a certain bit range of the pixel (or channel, in the case of PPM images). For example, if asked to plot the range 0-5, it will show the histogram (and corresponding entropies) of the six lower bits of each pixel (or channel).

2. Implement a program, named `img_quant`, to reduce the number of bits used to represent each image pixel/channel (i.e., to perform uniform scalar quantization). The result should also be an image file, in RAW, PGM or PPM format.

3. Implement a program, named `img_cmp`, that prints:

   - The maximum per pixel/channel absolute error.
   - The peak-signal-to-noise ratio (PSNR) of a certain image file in relation to its original version.

Part II

4. Develop a lossless image codec, with the best performance that you can obtain. Note that performance means not only compression ratio, but also compression/decompression speed and memory usage. The encoder should accept a RAW, PGM or PPM image file and produce the compressed version of it. The decoder should accept a compressed version and produce a RAW, PGM or PPM image file exactly equal to the original file. The calling format for PGM and PPM images should be

   ```
   img_lossless_enc [ optional flags ] input_img compressed_file
   img_lossless_dec compressed_file output_img
   ```

   and, for 8 bpp RAW images,

   ```
   img_lossless_enc [ optional flags ] -rows n_rows -cols n_cols
     input_img compressed_file
   img_lossless_dec compressed_file output_img
   ```

   and, for 16 bpp RAW images,

   ```
   img_lossless_enc [ optional flags ] -rows n_rows -cols n_cols
     -bpp 16 [-be|-le] input_img compressed_file
   img_lossless_dec compressed_file output_img
   ```

   where `[-be|-le]` indicate big-endian and little-endian, respectively.

5. Your codec should be compared with state-of-the-art lossless codeds, not only those specialized for image coding (e.g., `jpeg-ls`, `jpeg-xl`, ...), but also others that are general purpose (e.g., `zstd`, `lzma`, ...). This comparison should include the compression ratio (expressed in average number of bits per pixel), but also compression/decompression time and memory usage.

Part III

6. Develop a lossy image codec aiming at the best possible rate-distortion performance. All design decisions are left to the group and must be experimentally justified. The calling format should be identical to the one specified for the lossless case, with "`img_lossless_enc`" and "`img_lossless_dec`" changed to "`img_lossy_enc`" and "`img_lossy_dec`", respectively. However, an additional encoder flag should be considered, `-q quality`, to control the quality of the compressed image, where "`quality`" is an integer in the range 1-100.

7. Your codec should be compared with state-of-the-art lossy codeds, such as those of the JPEG family. This comparison should include the compression ratio (expressed in average number of bits per pixel) and the corresponding PSNR (i.e., the rate-distortion curve), but also compression/decompression time and memory usage.

Part IV

8. **Report 1**: Provide a report (12 pages at most), where you describe all the relevant steps and decisions taken in all the items of the work (for example, why a certain programming language was used to implement a certain part of the work; why did you choose a certain method; ...). When appropriate, include measures of processing time, compression ratios and corresponding errors introduced by the compression process (rate-distortion curves). For this, use several image files. This report should not include code. Instead, it should explain the main blocks of the system, why they were chosen and why do you think they are the most appropriate. Also, it should include a concise manual explaining how the software can be ran.

9. **Report 2**: Provide a report (6 pages at most), indicating all AI tools used to help you during the development of the work, together with a critical assessment of their effectiveness. It is expected that in this report you address questions such as: What was the problem? What did the AI propose? How did you verify it? Was it correct? What did you change? What did you learn?